

Rezolvarea sistemelor lineare

Sisteme liniare triunghiulare $\rightarrow$ *backward/forward substitution*

$$\left\{ \begin{array}{l} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots \dots \dots + a_{1n}x_n = b_1 \\ \quad a_{22}x_2 + a_{23}x_3 + \dots \dots \dots + a_{2n}x_n = b_2 \\ \quad \quad \ddots \\ \quad \quad \quad a_{ii}x_i + a_{ii+1}x_{i+1} + \dots \dots \dots + a_{in}x_n = b_i \\ \quad \quad \quad \quad \ddots \\ \quad \quad \quad \quad \quad a_{n-1\ n-1}x_{n-1} + a_{n-1\ n}x_n = b_{n-1} \\ \quad \quad \quad \quad \quad \quad a_{nn}x_n = b_n \end{array} \right.$$

Sisteme lineare compatibile determinate

$$\left\{ \begin{array}{l} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2 \\ \quad \quad \quad \vdots \\ a_{i1}x_1 + a_{i2}x_2 + \dots + a_{in}x_n = b_i \\ \quad \quad \quad \quad \quad \quad \vdots \\ a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n = b_n \end{array} \right. \Leftrightarrow \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \dots & \vdots \\ a_{i1} & a_{i2} & \dots & a_{in} \\ \vdots & \vdots & \dots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_i \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_i \\ \vdots \\ b_n \end{bmatrix}$$

$$A \text{ inversabilă } (\det A \neq 0), A \cdot x = b \Rightarrow x = A^{-1} \cdot b.$$

Eliminarea gaussiană naivă

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1k} & \dots & a_{1j} & \dots & a_{1n} \\ 0 & a_{22} & \dots & a_{2k} & \dots & a_{2j} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots & \dots & \vdots & \dots & \vdots \\ 0 & a_{k2} & \dots & a_{kk} & \dots & a_{kj} & \dots & a_{kn} \\ \vdots & \vdots & \dots & \vdots & \dots & \vdots & \dots & \vdots \\ 0 & a_{i2} & \dots & a_{ik} & \dots & a_{ij} & \dots & a_{in} \\ \vdots & \vdots & \dots & \vdots & \dots & \vdots & \dots & \vdots \\ 0 & a_{n2} & \dots & a_{nk} & \dots & a_{nj} & \dots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_k \\ \vdots \\ x_i \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_k \\ \vdots \\ b_i \\ \vdots \\ b_n \end{bmatrix}$$

$$[a_{11} \quad a_{12} \quad \dots \quad a_{1k} \quad \dots \quad a_{1j} \quad \dots \quad a_{1n}]$$

$$\begin{bmatrix}
 a_{11} & a_{12} & \dots & a_{1k} & \dots & a_{1j} & \dots & a_{1n} \\
 0 & a_{22} & \dots & a_{2k} & \dots & a_{2j} & \dots & a_{2n} \\
 \vdots & \vdots & \ddots & \vdots & \dots & \vdots & \dots & \vdots \\
 0 & 0 & \dots & a_{kk} & \dots & a_{kj} & \dots & a_{kn} \\
 \vdots & \vdots & \dots & \vdots & \dots & \vdots & \dots & \vdots \\
 0 & 0 & \dots & a_{ik} & \dots & a_{ij} & \dots & a_{in} \\
 \vdots & \vdots & \dots & \vdots & \dots & \vdots & \dots & \vdots \\
 0 & 0 & \dots & a_{nk} & \dots & a_{nj} & \dots & a_{nn}
 \end{bmatrix}
 \begin{bmatrix}
 x_1 \\
 x_2 \\
 \vdots \\
 x_k \\
 \vdots \\
 x_i \\
 \vdots \\
 x_n
 \end{bmatrix}
 =
 \begin{bmatrix}
 b_1 \\
 b_2 \\
 \vdots \\
 b_k \\
 \vdots \\
 b_i \\
 \vdots \\
 b_n
 \end{bmatrix}$$

$$\begin{bmatrix}
 a_{11} & a_{12} & \dots & a_{1k} & \dots & a_{1j} & \dots & a_{1n} \\
 0 & a_{22} & \dots & a_{2k} & \dots & a_{2j} & \dots & a_{2n} \\
 \vdots & \vdots & \ddots & \vdots & \dots & \vdots & \dots & \vdots \\
 0 & 0 & \dots & a_{kk} & \dots & a_{kj} & \dots & a_{kn} \\
 \vdots & \vdots & \dots & \vdots & \ddots & \vdots & \dots & \vdots \\
 0 & 0 & \dots & 0 & \dots & a_{jj} & \dots & a_{jn} \\
 \vdots & \vdots & \dots & \vdots & \dots & \vdots & \dots & \vdots \\
 0 & 0 & \dots & 0 & \dots & 0 & \dots & a_{nn}
 \end{bmatrix}
 \begin{bmatrix}
 x_1 \\
 x_2 \\
 \vdots \\
 x_k \\
 \vdots \\
 x_j \\
 \vdots \\
 x_n
 \end{bmatrix}
 =
 \begin{bmatrix}
 b_1 \\
 b_2 \\
 \vdots \\
 b_k \\
 \vdots \\
 b_i \\
 \vdots \\
 b_n
 \end{bmatrix}$$

Exemplu (în care eliminarea gaussiană naivă nu funcționează):

$$\begin{cases} \varepsilon x_1 + x_2 = 1 \\ x_1 + x_2 = 2 \end{cases} \Rightarrow \begin{cases} \varepsilon x_1 + x_2 = 1 \\ (1 - \varepsilon^{-1})x_2 = 2 - \varepsilon^{-1} \end{cases} \Rightarrow x_2 = \frac{2 - \varepsilon^{-1}}{1 - \varepsilon^{-1}} \approx 1, x_1 = \frac{1 - x_2}{\varepsilon} \approx 0$$

$$\begin{cases} x_1 + x_2 = 2 \\ \varepsilon x_1 + x_2 = 1 \end{cases} \Rightarrow \begin{cases} x_1 + x_2 = 2 \\ (1 - \varepsilon)x_2 = 1 - 2\varepsilon \end{cases} \Rightarrow x_2 = \frac{1 - 2\varepsilon}{1 - \varepsilon} \approx 1, x_1 = 2 - x_2 \approx 1$$

Descompunerea L,U,P și alte descompuneri

$$L = \begin{bmatrix} 1 & & & & \\ \ell_{21} & 1 & & & \\ \ell_{31} & \ell_{32} & 1 & & \\ \vdots & \vdots & \vdots & \ddots & \\ \ell_{n1} & \ell_{n2} & \ell_{n3} & \cdots & 1 \end{bmatrix} \quad U = \begin{bmatrix} u_{11} & u_{12} & u_{13} & \cdots & u_{1n} \\ & u_{22} & u_{23} & \cdots & u_{2n} \\ & & u_{33} & \cdots & u_{3n} \\ & 0 & & \ddots & \vdots \\ & & & & u_{nn} \end{bmatrix}$$

$P = I_n$ cu liniile permutate

★ Descompunerea L,U,P: $P \cdot A = L \cdot U$

💡 Rezolvarea sistemului $A \cdot x = b$ se reduce la rez. a două sisteme triunghiulare:

$$A \cdot x = b \Leftrightarrow (P \cdot A) \cdot x = P \cdot b \Leftrightarrow L \cdot U \cdot x = P \cdot b \Leftrightarrow L \cdot y = P \cdot b, U \cdot x = y$$

! Descompunerea L, U, P a matricei A se poate folosi pentru b-uri diferite în rezolvarea sistemului $A \cdot x = b \Rightarrow$ economie de memorie și timp de execuție.

★ Dacă A este **simetrică pozitiv definită**, atunci putem obține descompunerea Cholesky ($L = R^*, U = R, P = I_n$):

$$A = R^* \cdot R, \text{ unde } R \text{ este triunghiulară superior.}$$